



Semester I Examinations 2019/20

Exam Code(s) 1SD1, 1SD3, 1MF1
Exam(s) Higher Diploma (Software Design & Development)
M.Sc. (Software Design & Development)
Higher Diploma (Software Design & Development) –
Industry Stream

Module Code(s) CT516
CT861
CT538
CT542

Module(s) Computer Architecture and Operating Systems

Paper No. 1
Repeat Paper No

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Instructions: Answer **question 1** and **two** other questions.

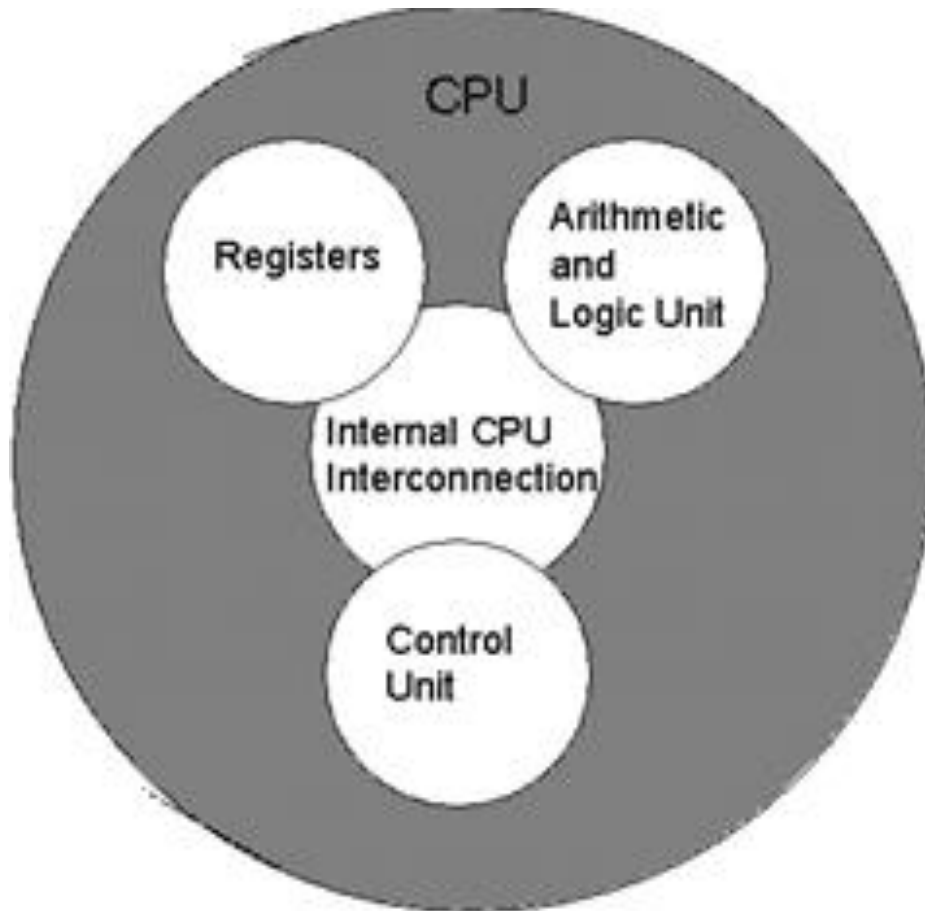
Duration 2 hrs
No. of Pages 4
Discipline(s) School of Computer Science

Requirements None

Release to Library: Yes ☒ No ☐

Question 1 (COMPULSARY)

- a) Using examples explain the purpose / functionality of the four main structural components of a CPU as shown in the diagram below. Further on, explain the interaction of these components when executing an instruction. In your answer refer to all important register types.



7 MARKS

- b) Using an example outline in detail how the *paged memory allocation* scheme works. In your answer refer to the terms, *pages*, *frames*, *page tables*, and *demand paging*.

7 MARKS

- c) Discuss the purpose and inner workings of the following protocols:
- Dynamic Host Configuration Protocol (DHCP)
 - Address Resolution Protocol (ARP)

Use diagrams to support your answer.

6 MARKS

PTO

Question 2

- a) What is a *half-adder*? Provide its truth table, therefore showing how it can be implemented using logic gates.
6 MARKS
- b) What is meant by the *von Neuman bottleneck*? Using diagrams explain how the Harvard architecture or the use of caches alleviates this problem.
4 MARKS
- c) Briefly discuss the importance of the *medium access sublayer* and distinguish between the following *multiple access protocols*:
- 1-persistent CSMA
- non-persistent CSMA
- p-persistent CSMA
5 MARKS
- d) Show how an “*if-then-else*” block in a high-level programming language such as Java may be translated into a series of CPU instructions. In your answer explain the concept of the *conditional jump* instruction and the *unconditional jump* instruction.
5 MARKS

Question 3

- a) Differentiate between the logic gates *AND*, *OR*, *XOR*, and *NAND*. Draw their truth tables, assuming that each gate has two inputs A and B.
4 MARKS
- b) Summarise the purpose and functionality of the different hardware components you would typically find in a local area network (LAN).
5 MARKS
- c) Using machine instruction examples and diagrams distinguish between the following addressing modes:
• Immediate Addressing
• Direct Addressing
4 MARKS
- d) Summarise the *7-state process model* and, using an example, show how a process moves through all different states, hereby clearly outlining the reason for each state change.
7 MARKS

PTO

Question 4

- a) Using the code fragment below, explain the term *locality of reference*:

```
int a = 0, b = 1;
for (a = 1; a < 10000; a++) {
    b = b * a;
}
```

Why is *locality of reference* an important pre-requisite for the efficient use of caches?

4 MARKS

- b) Distinguish between the following cache-related techniques, clearly outlining their purpose (and possible advantages) when operating a cache:

- *Write back* and *write through*
- *Direct mapping*, *associative mapping* and *set associative mapping*
- *Least recently used* and *first in first out* replacement

5 MARKS

- c) Explain the inner workings of a hard disk with multiple platters. In your answer refer to the terms *sector*, *track* and *cylinder*.

5 MARKS

- d) Summarise the functionality of the various layers of the TCP/IP protocol stack. Using a diagram show the overall structure of a packet (consisting of various headers and the payload) when it eventually “hits the wire”, i.e. enters a local area network.

6 MARKS